Reg. No. | | | | | | | | | | | [24MTVES101]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY 2025
LOW POWER VLSI DESIGN

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

		Questions	CO's	Blooms Level	Marks
1.	a.	Draw a NMOS transistor structure and explain all the three region of operation with current equation.	CO1	L2	10
	b.	Draw a CMOS Inverter transfer characteristics and explain each region of operation as input vary from 0 to V_{dd}.	CO1	L2	10
2.	a.	Briefly explain the need of low power VLSI design. With proper diagram and equation explain the sources of power dissipation.	CO2	L2	10
	b.	With a help of CMOS inverter logic, explain the condition for which short circuit current flows in it. And briefly explain methods to reduce the short circuit power.	CO2	L3	10
3.	a.	Explain the VLSI design flow and simulation tool used at each level of abstraction.	CO3	L2	10
	b.	Explain the emerging low power approaches used in the CMOS.	CO3	L2	10
		OR			
4.	a.	Explain the different recurring themes in low power design with suitable example.	CO3	L2	10
	b.	With a help of vector quantization example, show that tree search algorithm reduces the dynamic power dissipation.	CO3	L3	10
5.	a.	With a necessary equation and graph, describe the effect of V_{dd} and V_t on speed.	CO4	L2	10
	b.	What must be the transistor sizing and optimal gate oxide thickness for low power design?	CO4	L3	10

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		OR			
6.	**a.**	Draw the structure of Silicon on Insulator (SoI), and mention the advantages and disadvantages of SoI over CMOS.	**CO4**	**L2**	**10**
	b.	With neat diagram, explain the concept of DTCMOS low power technique.	**CO4**	**L2**	**10**
7.	**a.**	What are the different types of Flip-flops, estimate the total power dissipation of transmission gate based static flip-flop.	**CO5**	**L3**	**10**
	b.	With suitable diagram, explain Bus Invert Encoding logic used for minimizing the power during data transmission.	**CO5**	**L3**	**10**
		OR			
8.	**a.**	Apply the Precomputation low power technique to 4 bit comparator (A>B) and explain the same with neat block diagram.	**CO6**	**L4**	**10**
	b.	Explain the concept of Signal gating, logic encoding technique used to reduce the switching activity in the circuit.	**CO6**	**L3**	**10**